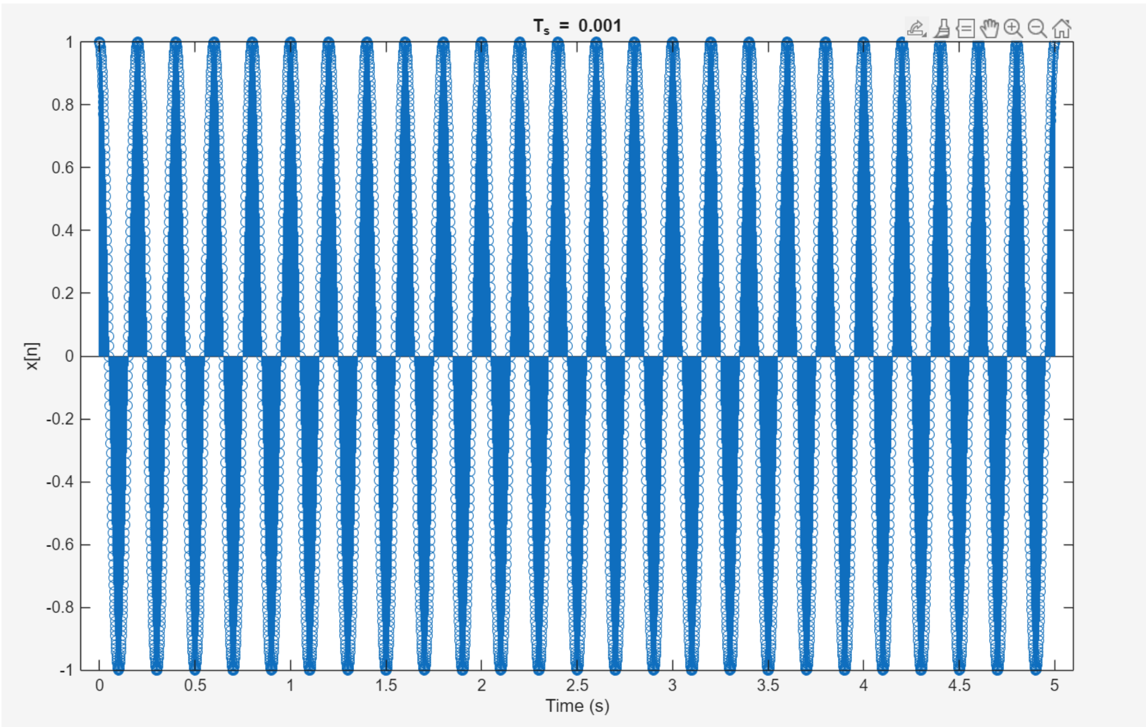
EE110B Lab#7

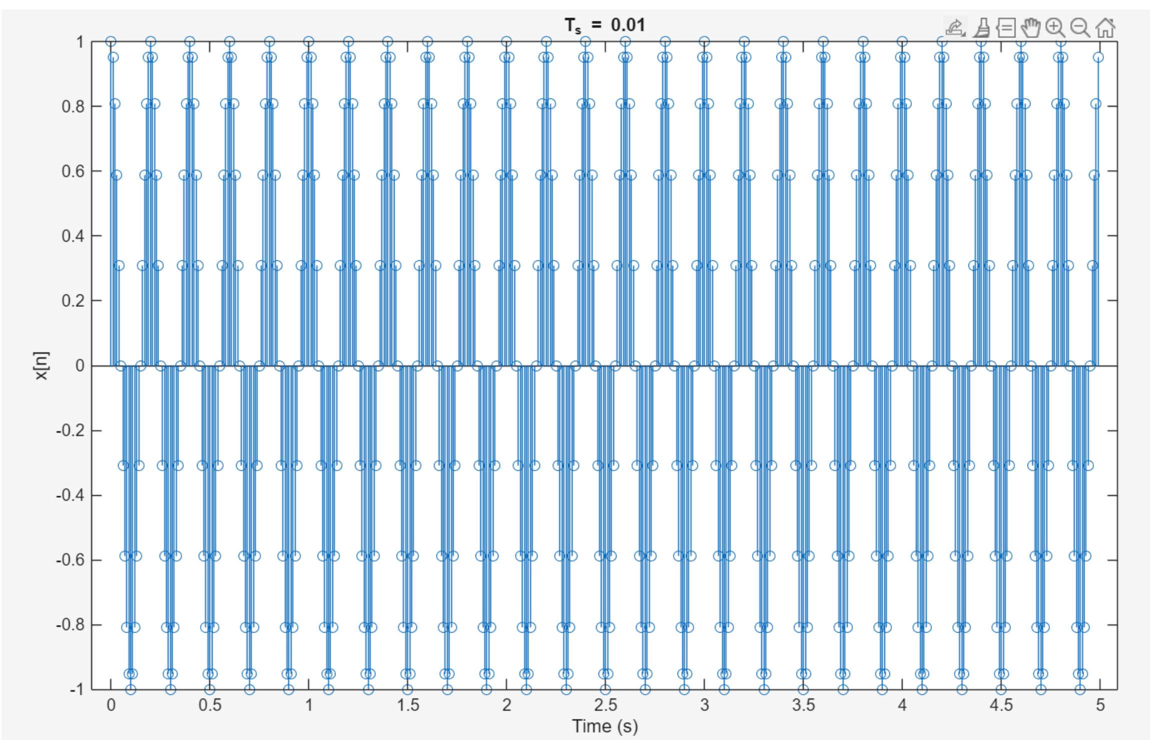
Nathan Taing

Aliasing of Sinusoids

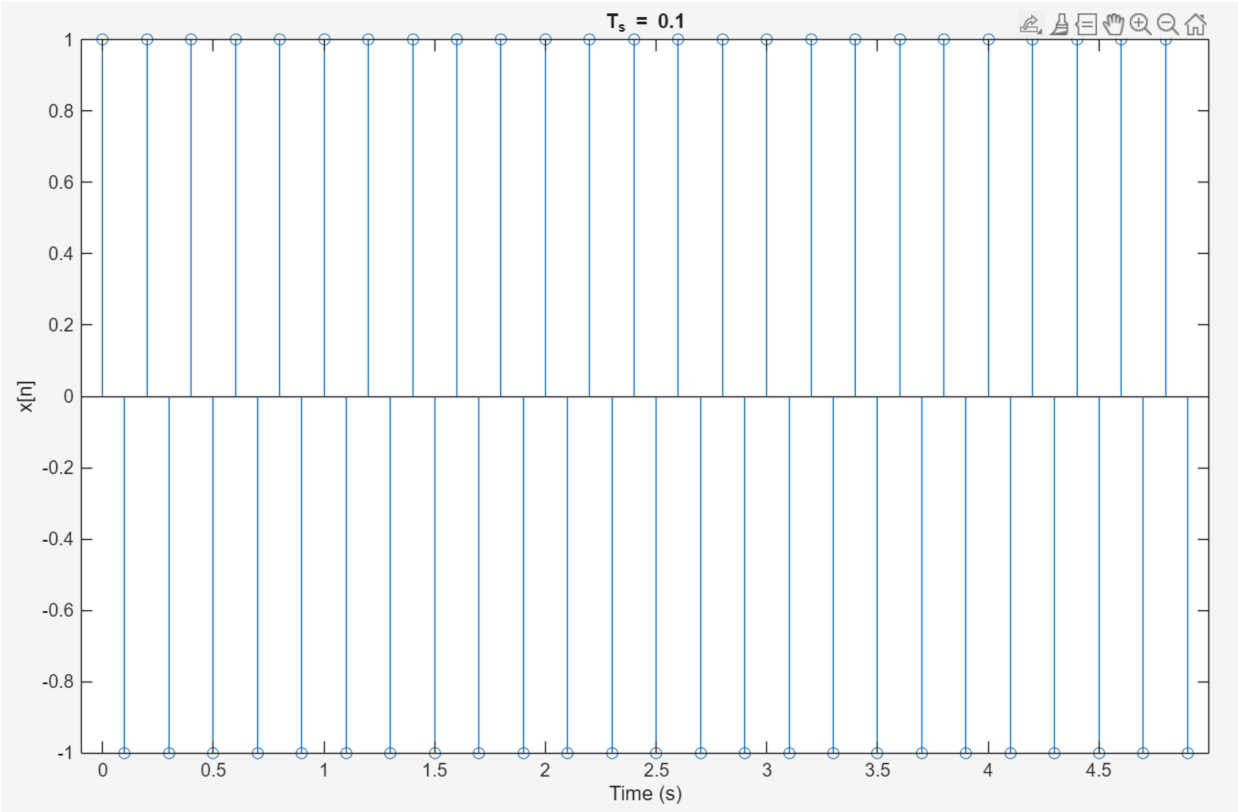
a) Sampling period: 0.001s



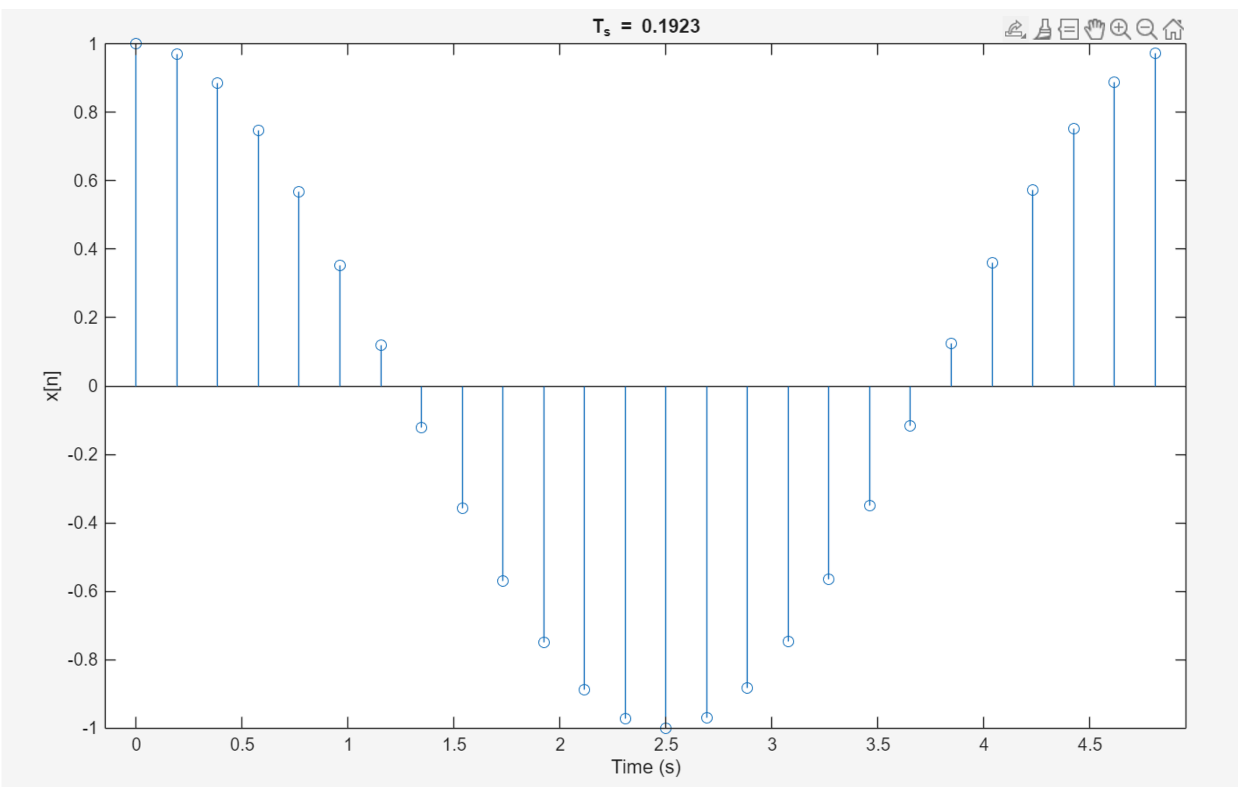
b) Sampling period: 0.01s



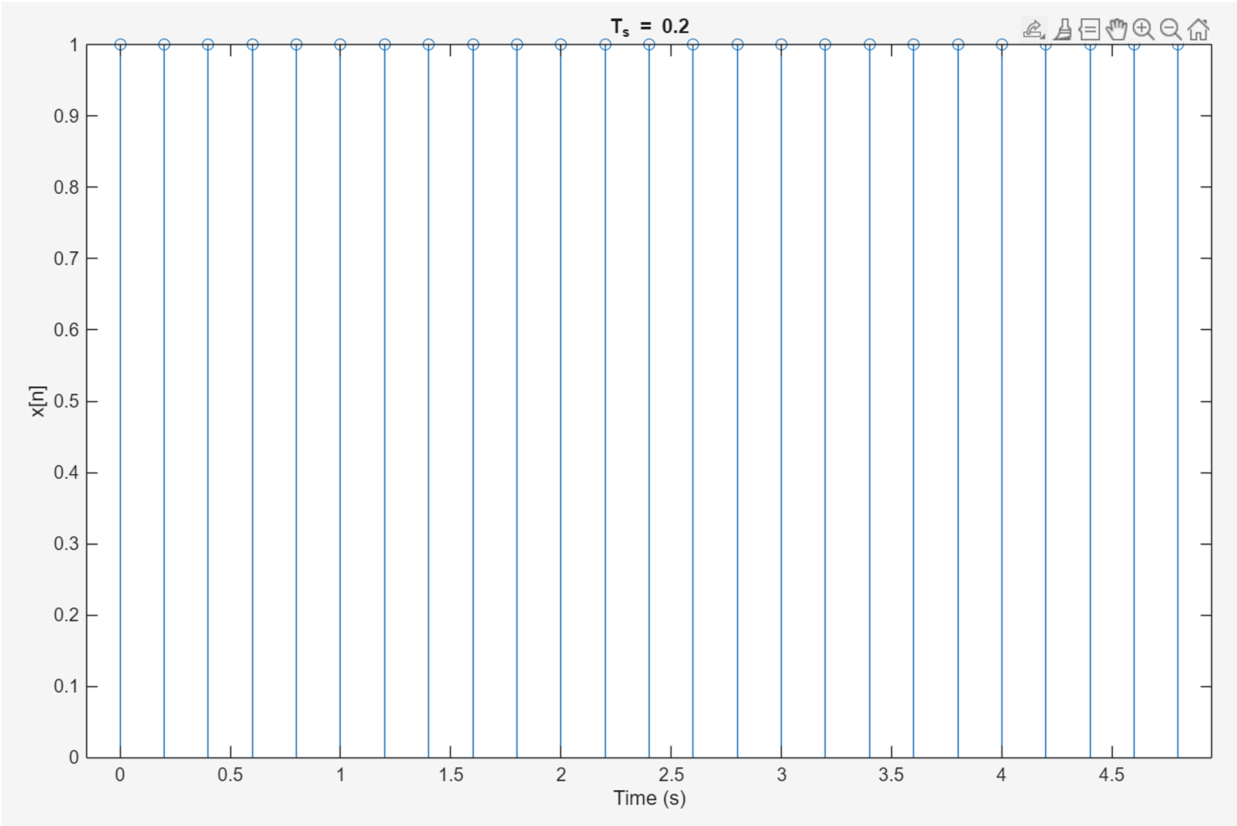
c) Sampling period: 0.1s



d) Sampling period: 0.1923s



e) Sampling period: 0.2s



Code:

```
Ts_values = [0.001 0.01 0.1 0.1923 0.2];
  k = 5;                    %change value depending on which value we are testing
  Ts = Ts_values(k);
  n = 0:(5/Ts)-1;
  x = cos(10*pi*Ts*n);
  figure
  stem(n*Ts, x)
  title(['T_s = ', num2str(Ts)])
  xlabel('Time (s)')
  ylabel('x[n]')
```

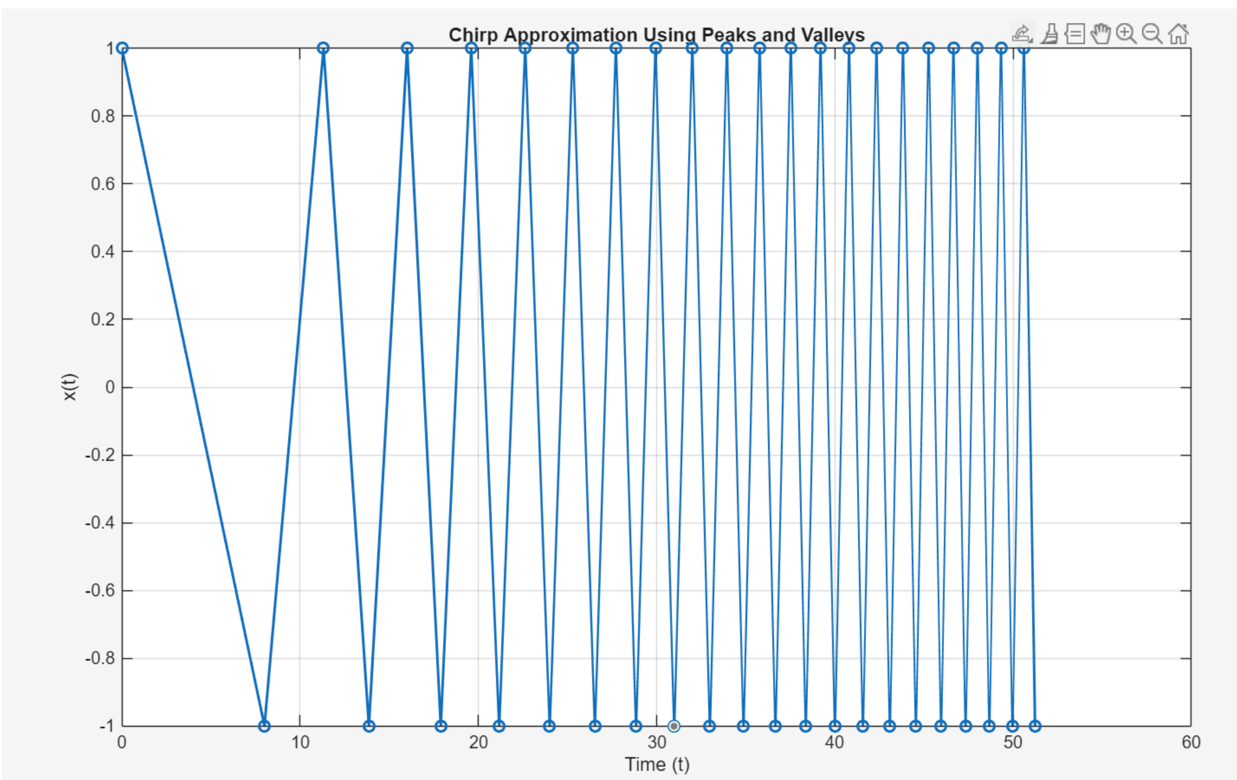
With smaller sampling periods, the signal can plot all 25 cycles. We can see the oscillation that x[n] represents. However, when we reach the sampling period of 0.1932s, x[n] does not plot 25 cycles. It only plots one cycle. When we reach the sampling period of 0.2s, it becomes a constant function where all values of n become one.

This behavior can be explained by aliasing. Our original x[n] signal has a frequency of 5 Hz. The Nyquist criterion says that Ts < 0.1 is the largest allowable sampling period to acquire samples that maintain the original x[n] shape. When we plug in the sampling period of 0.1932, we get approximately a sampling frequency 5.2 Hz. The sampling rate is insufficient to properly detail the original x[n] signal, as it is longer than the base frequency. This results in a sampled x[n] signal that takes on a new shape. The sampling is too slow, which results in our graph looking like a completely different graph than before. It appears x[n] only performs 1 oscillation in part d, which is not what is actually happening. For part e, the sampling period exactly equals the original period of x[n]. This results in the sampler sampling the same point at every time interval. As a result, this sampled x[n] signal looks like it holds a constant value. This is also due to aliasing.

Aliasing of Chirps

a) No, x(t) is not periodic because as t increases, the frequency of the signal increases with it. Additionally, the phase contains t squared, which means the oscillations grow quickly, The space from peak to peak decreases as there is no constant T to produce a constant period length.

b) Connecting $(a_k, 1)$ and $(\beta_k, -1)$



Code:

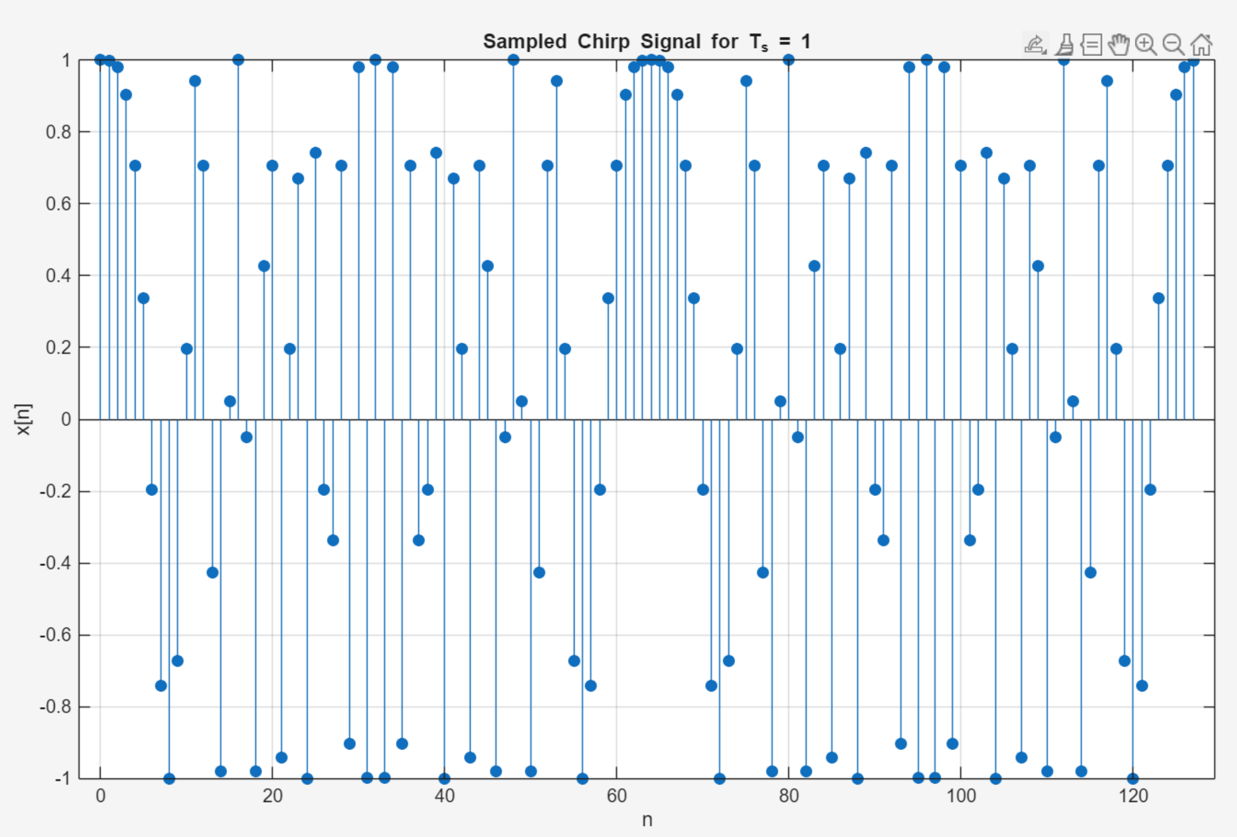
```
k = 0:20;

alpha_k = sqrt(128 * k);
```

```
beta_k = sqrt(64 * (2 * k + 1));
t_points = zeros(1, 2*length(k));
x_points = zeros(1, 2*length(k));
for i = 1:length(k)
  t_points(2*i - 1) = alpha_k(i);
  x_points(2*i - 1) = 1;
  t_points(2*i) = beta_k(i);
  x_points(2*i) = -1;
end
figure;
plot(t_points, x_points, '-o', 'LineWidth', 1.5);
xlabel('Time (t)');
ylabel('x(t)');
title('Chirp Approximation Using Peaks and Valleys');
grid on;
```

The distance between the consecutive crests shrinks as time increases. This matches the behavior of the chirp because we are expecting the frequency to increase with time. If we look at the x[n] signal, we will see the peak to peak of the x[n] getting closer and closer.

c)

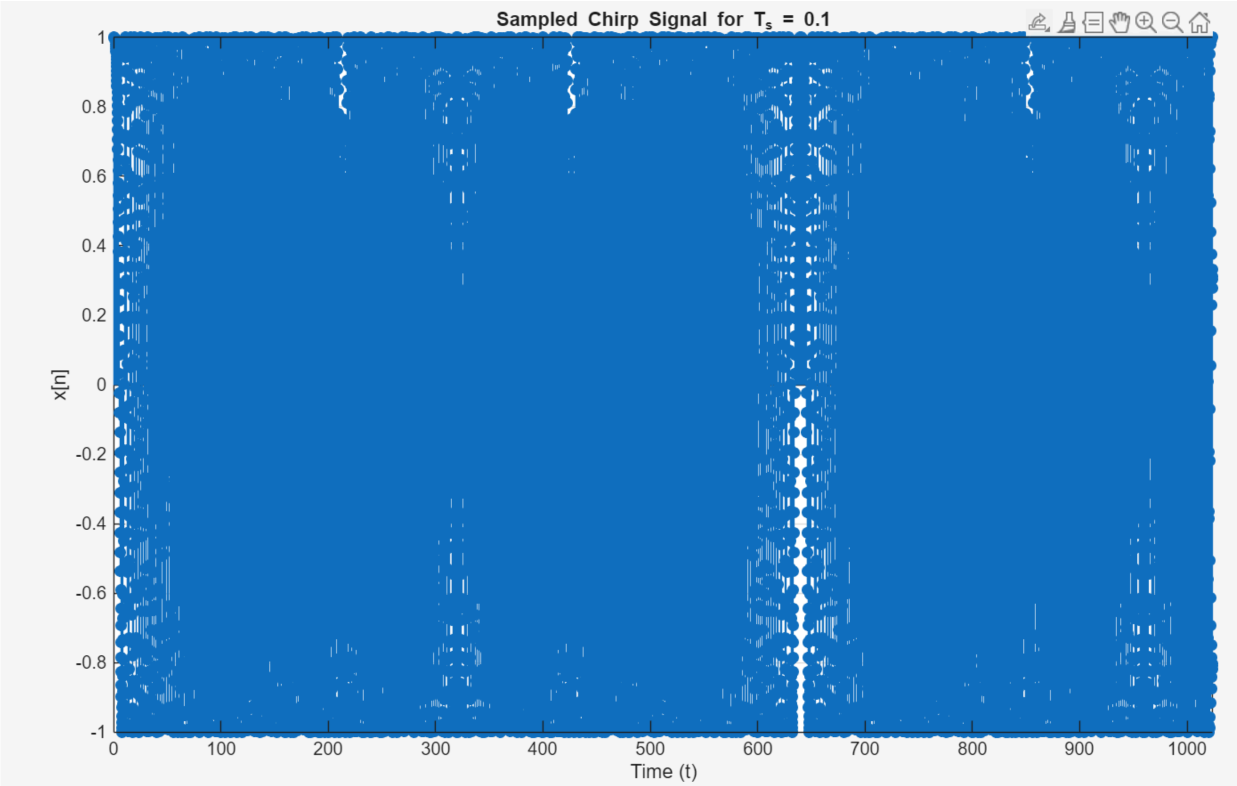


Code:

```
Ts = 1;
n = 0:(128/Ts)-1;
x = cos((pi * Ts^2 * n.^2) / 64);
figure;
stem(n, x, 'filled');
xlabel('n');
ylabel('x[n]');
title('Sampled Chirp Signal for T_s = 1');
grid on;
```

It appears, in this sample of x[n], the signal becomes periodic. This is different from before as x(t) does not have periodic behavior. This can be attributed to aliasing and phase wrapping. Cosine functions repeat every (2/pi), resulting in this sampled x[n] appearing to have a periodic pattern. We can note that sampled signals behave differently from the original continuous-time chirp. The use of sampling can cause the appearance of x[n] resulting in signals with behaviors different from the original signal we are sampling from.

d)



Code:

```
Ts = 0.1;
n = 0:(128/Ts)-1;
x = cos((pi * Ts^2 * n.^2) / 64);
figure;
stem(n, x, 'filled');
xlabel('n');
ylabel('x[n]');
title('Sampled Chirp Signal for T_s = 0.1');
grid on;
```

Utilizing a smaller sample period resulted in a closer sample signal to part b. We can see the signal starting off slowly, and then increasing as time increases. We should note this is not a universal solution from the signal over larger time intervals. This is because we can see some aliasing behavior around t = 640. From t = 0 to around t = 600, we can see the frequency of the chirp increasing. However, the Nyquist criterion is around t = 640, meaning anything higher than this frequency will result in the sampled signal appearing distorted and exhibiting behaviors different from the original signal.

e)

No, we cannot choose a fixed sampling period that works for every interval of t. The behavior of the chirp is that the frequency will increase as time increases. If we use a very small sampling period, the signal frequency will eventually increase high enough to exceed the Nyquist criterion. This means aliasing is inevitable if we plot the signal over a great enough number of time. If we had a different original signal, we could choose a fixed sampling period, however, this is not a signal that can exhibit this behavior.